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A data-driven digital twin model for bridge health monitoring using feature fusion and unsupervised deep learning

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ABSTRACT

The progressive degradation of bridge infrastructure poses significant risks to transportation safety, necessitating advanced monitoring and management strategies. Digital Twins (DTs) have emerged as a transformative technology for real-time structural monitoring, offering a dynamic and data-driven representation of physical assets. When combined with Deep Learning (DL), DTs enhance predictive capabilities by leveraging vast amounts of sensor data to detect structural anomalies, optimize maintenance strategies, and improve decision-making processes. This synergy enables a more accurate, automated, and cost-effective approach to infrastructure health monitoring, ensuring timely interventions and prolonged service life. This paper presents a novel data-driven DT framework for real-time bridge damage detection, incorporating two key innovations. First, a Multi-Domain Fusion Autoencoder (MDFA) that integrates raw acceleration data with feature fusion from time, frequency, and wavelet domains. Second, a synthetic anomaly simulation approach for evaluation. The framework employs a transformer-based Generative Adversarial Network (GAN) to generate realistic synthetic signals, which are modified to simulate various damage scenarios. Evaluations on Australia's Werrington Bridge demonstrate the MDFA's ability to detect subtle anomalies often overlooked by conventional unsupervised DL methods. A comparative analysis confirms its superior detection capabilities. Further validation on Austria's S101 Bridge using real damage data successfully identifies and localizes all damage scenarios without requiring additional optimization. These results highlight the efficiency of feature fusion techniques in improving anomaly detection accuracy and suggest that synthetic anomaly simulation offers a cost-effective alternative to laboratory testing for evaluating unsupervised damage detection approaches.

1. Introduction

Bridge infrastructures, as a critical component of the transportation network, are experiencing significant deterioration due to increasing traffic volume and service load, ageing, and the impacts of aggressive environmental factors associated with climate change. The progressive deterioration of bridge elements results in extensive damage, leading to significantly increased repair or replacement costs. If left unaddressed, it may ultimately cause catastrophic structural failure, resulting in severe economic losses and, tragically, the loss of human lives. Therefore, the proper inspection and maintenance of bridge infrastructure through Structural Health Monitoring (SHM) approaches have become increasingly important for ensuring structural reliability and safety [1–3].

SHM systems play a crucial role in ensuring the safety and reliability of bridge infrastructures by detecting damage and evaluating structural conditions. SHM approaches can be broadly categorized into model-driven and data-driven methods. Model-driven approaches rely on physics-based models and finite element analysis to predict structural behaviour, while data-driven methods leverage sensor data and statistical techniques to identify patterns indicative of deterioration [4,5]. The increasing complexity of structural systems and the vast amount of monitoring data have led to a growing reliance on data-driven methods, particularly those incorporating Machine Learning (ML) techniques.

ML techniques have significantly transformed SHM by enhancing the ability to analyse large-scale monitoring data and have become a cornerstone of modern SHM systems, offering powerful tools for

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detecting and diagnosing structural damages. Previous studies have developed approaches to detect damage in bridges using both supervised and unsupervised ML techniques [6–8]. Supervised ML studies have demonstrated strong performance in controlled scenarios; however, their practical applicability to real-world cases remains limited [9,10]. In contrast, unsupervised learning provides a flexible and adaptive method for extracting valuable insights, as it operates on unlabelled data. Its ability to dynamically adjust to variations in data distributions and detect emerging trends makes it efficient for real-time monitoring applications. Several research studies have explored unsupervised techniques for the real time analysis of structural monitoring data. Despite the extensive progress made in unsupervised damage detection in bridges, the extraction of meaningful results from extensive real-time measurement data is a significant challenge, which affects the timely assessment of structural conditions and the implementation of appropriate remedial measures [9].

Digital Twins (DTs), as a powerful tool for online monitoring and predictive maintenance, have gained significant popularity in bridge infrastructure management. Various approaches, such as big data analysis, ML, and cloud/edge computing can be employed in DT platforms to manage the increasing volume of input data, and facilitate continuous, and ideally real-time, monitoring, which is crucial in identifying defects or problems at an early stage [11]. By analysing the collected data, DTs can identify potential issues within a system, enabling bridge management teams to make informed predictive maintenance decisions. This enhances reliability, increases efficiency, and optimizes the use of available resources [12]. The development of DTs mechanisms within Bridge Management Systems (BMS) in recent years has provided valuable opportunities to address real time monitoring challenge, particularly through advancements in applying ML algorithms within DTs framework [13].

A DT model serves as a dynamic virtual representation of physical infrastructure, providing a real-time reflection of its behaviour, current condition, and performance. This synchronization between the virtual and physical entities is facilitated by advanced systems with high-speed communication capabilities. These intelligent systems, encompassing advanced technologies such as Deep Learning (DL) and data analytics, enable computers to execute tasks that usually demand human intelligence. They achieve this by collecting and analysing large volumes of data from IoT-enabled (Internet of Things) data collection methods and other sensors, to identify patterns in data structure and make predictions with minimal human oversight [14].

Data-driven simulation models in DTs are developed based on the validated data obtained from experiments/measurements on the actual structure and produce a model to analyse the status of the bridge in real time. In data-driven models, data from the physical bridge is collected using emerging sensor technologies and IoT devices. This data is then processed using Big Data analytics and advanced ML techniques, to simulate the exact functionality of the bridge [15]. Consequently, bridge DT is equipped with dynamic ML-based data-driven models that continuously updates using real-time data analysis enabling the damage detection in real time.

While significant advancements have been made in developing of different virtual models within bridge DTs, such as geometric modelling via unmanned aerial vehicles [16,17] and terrestrial laser scanning [18,19], as well as Finite Element Method (FEM) models for numerical analysis [20,21], the incorporation of robust data-driven DT frameworks for long-term monitoring remains a challenge. Current DT implementations often lack comprehensive data-driven models for processing long-term bridge data [11,22,23].

Additionally, the use of real measured data and unsupervised DL methods for real-time damage detection in bridges still remains underexplored [24,25]. A key challenge is the lack of historical damage data in existing operational bridges. While laboratory and benchmark structures provide controlled environments for testing, they do not fully represent real-world complexities due to differences in scale and dynamic behaviour, environmental conditions, and operational variability.

Data fusion is a well-known technique in SHM, enabling the integration of heterogeneous data sources to enhance damage detection and structural assessment. By combining multiple data modalities, data fusion improves the reliability and accuracy of monitoring systems. Feature fusion, as a specific type of data fusion, plays a critical role in extracting meaningful insights from complex bridge data using real measured data and unsupervised DL techniques. Given the complexity of bridge dynamics, relying on a single domain limits the ability to capture intricate damage signatures in data-driven models within DTs frameworks [26,27].This limitation hinders the ability to detect structural deterioration promptly, leading to increased maintenance costs and potential safety risks.

This study introduces an advanced data-driven DT model for real-time bridge damage detection, leveraging feature fusion and novel unsupervised DL techniques. The proposed AI-enabled DT employs a Multi-Domain Fusion Auto encoder (MDFA) to integrate complementary features extracted from time, frequency, and wavelet domains to provide unique insights into structural responses. This framework enhances the DT applications for robust, real-time analysis and damage detection, owing to its capabilities to integrate extensive amount of sensor data which provide a comprehensive understanding of the structure's behaviour, performance, and status [11,28]. Moreover, the real time data processing provides the opportunity for early detection of structural issues, enabling timely interventions to prevent failures [29] and implement proactive maintenance strategies to extend the structure's life, reduce downtime, and lower maintenance costs [30].

The primary contributions and novelties of this study include:

1. Introducing a data-driven DT framework with a real-time data analysis and processing modules to prolong service life of bridges, advancing BMS toward faster and adaptable performance.
2. Integrating feature fusion techniques with DL within DT framework, which leverages complementary strengths to enhance real-time damage detection through adaptive unsupervised learning on real-world bridge data.
3. Developing a novel MDFA model that integrates raw acceleration data with features from time, frequency, and wavelet domains, significantly enhancing the detection of subtle structural anomalies often missed by conventional methods.
4. Proposing an innovative synthetic anomaly simulation approach using transformer-based Generative Adversarial Networks (GANs) to generate realistic synthetic signals for evaluation, providing a cost-effective alternative to laboratory testing for unsupervised damage detection methods.
5. Comparison of the performance with the state-of-the-art unsupervised deep learning methods in the literature.
6. Applying real-world bridge case studies for real-time damage detection, including simulating new data points and alert generation.

Several experimental tests have been carried out on a real bridge case study. Results show that the proposed framework significantly enhances early damage detection and predictive maintenance capabilities, demonstrating its potential to revolutionize bridge health monitoring and ensure long-term infrastructure safety.

The outline of this paper is as follows: Section 2 provides the related work and outlines the motivation and objectives of the proposed approach. Section 3 presents the proposed solution with a focus on feature extraction and fusion, and the MDFA model. Section 4 describes the complete evaluation strategy including synthetic data generation, anomaly scenarios and applied metrics. Section 5 reports the results of the proposed framework on the first real bridge case study. Section 6 compares the proposed approach with an existing well-preforming method. In Section 7 the results of the second case study were discussed. Finally, Section 8 concludes the paper with final remarks and

suggestions for future work.

2. Related works and motivations

DL models have proven to be efficient in real-time SHM applications, enabling to process streaming data from different sensors. DL models automatically extract features through their hidden layers, and typically demand large volumes of data to perform effectively. Bayane et al. [9] proposed an approach for real-time detection of damage in bridges by leveraging unsupervised anomaly detection algorithms applied to strain and acceleration measurements. Li et al. [31] proposed an unsupervised damage detection method designed for real-time application during vehicle crossings over a bridge. Teng et al. [24] proposed an unsupervised damage detection method that employs an attention-based convolutional auto encoder to detect progressive damage in bridges. Kang et al. [32] proposed a DL-based anomaly detection method has been developed for both historical data mining and real-time alerting. Fernandez et al. [33] also employed a hybrid methodology that incorporates Finite Element simulations to enrich the training phase of a DL with synthetic damage scenarios for damage identification in bridge structures.

The rapid evolution of infrastructure management has led to the adoption DL methods within DT framework for efficient bridge health monitoring. The integration of DL algorithms has further advanced the capabilities of DTs by identifying hidden patterns within the large volumes of data and automating anomaly detection, predictive analytics, and model calibration. This capability allows for the application of AI-enabled DTs in different tasks including railway maintenance activities [34,35], building predictive maintenance [36] and smart manufacturing and advanced robotics[37].

DL methods have also been utilized for synthetic data generation to support anomaly detection in bridge DTs, particularly when real-world datasets are limited or incomplete. Jiménez Rios et al. [38] proposed a synthetic data generation framework to enhance the training of ML models for damage detection in bridge structures. In another research from the same authors, Jiménez Rios et al. [39] examined the uncertainties within synthetic data generation frameworks and the potential application of these methodologies as efficient and cost-effective alternatives to real monitoring, supporting the rapid development DTs. Similarly, Mousavi et al. [40] proposed a data augmentation approach to generate synthetic acceleration data in bridge DTs, aiming to mitigate the challenges of substantial training data requirements in DL models, which are often costly and time-intensive to collect. However, Limited research has explored the integration of DL methods within DTs frameworks for the real time analysis of bridge monitoring data. Table 1 presents the comparison of AI-enabled DT solutions (or related frameworks) for real-time bridge health monitoring as reported in recent literature.

The gaps in these methods include real-time performance constraints, technical scalability and optimized DL techniques. This study aims to address these key challenges through feature fusion, advanced pre-processing techniques, and optimized DL architectures to develop robust and efficient SHM system capable of managing the complexities of real-world environments.

3. Proposed data-driven DT framework

The integration of data analytics and DL within digital twinning enhances bridge monitoring and maintenance by enabling real-time structural assessment and predictive decision-making. To fully exploit these advantages, a data-driven reference framework is essential.

The proposed framework, depicted in Fig. 1, presents a layered architecture for efficient data analytics in bridge DTs. The process begins with sensor networks collecting real-time structural data. While this study specifically focuses on vibration data, the proposed data-driven framework is designed to be adaptable and can be extended to incorporate other structural parameters such as strain and temperature. This data is then transmitted through communication technologies and enters the data analysis layer, where different Artificial Intelligence models, statistical analyses, and mathematical approaches construct a data-driven digital representation of the bridge. The Artificial Intelligence section of the framework also incorporates other data-driven techniques, including synthetic data generation, which is used to generate realistic signals for structural health assessment in our workflow implementation (Section 5.1). To ensure structural integrity and safety, raw sensor data undergoes pre-processing, such as data cleaning, filtering, and feature extraction, before analysis using DL and feature fusion techniques. This enables bridge behaviour simulation, structural health assessment (including damage localization and severity estimation), and visualization to support decision-making. Ultimately, this data-driven digital twin framework facilitates proactive maintenance, improves operational efficiency, and extends bridge service life by providing actionable insights.

Table 1
Comparison of AI-enabled DT methods or frameworks for bridge health monitoring.

Approach	Description	Limitations	Ref
Cloud-Based DT with DL	Uses IoT sensors for strain, vibration, and environmental data. Implements DL for anomaly detection. Integrates real-time cloud analytics for dynamic DT updates.	1. Using simplified supervised 1D CNN model 2. Requires stable cloud connectivity.	[41]
ML-based DT model for SHM	Uses fiber-optic sensor and deep neural networks to convert strain output into an interactive DT visualization	1. Using simple ML model to generate the relevant baseline data. 2. Requires accurate and detailed FEM models 3. Not applicable to all type of bridge structures	[42]
DT and support vector machine (SVM) in SHM of bridges	ML and DT model are merged to predict damage or collapse of the bridge	1. Using supervised SVM model 2. Real-time performance constraints	[43]
DT and Transfer learning for bridge damage detection	A Large number of damage samples of the numerical models were obtained by using the DT and used to train a CNN as a pre-trained network. Transfer learning was used to test on a real bridge.	1. Implementation complexity may be high 2. Not applicable to all type of bridge damages 3. Real-time performance constraints	[44]

4. Workflow of bridge health monitoring via data-driven DT framework

Fig. 2 illustrates the detailed workflow of the proposed data-driven DT framework for real-time bridge health monitoring. The framework consists of four main phases: data acquisition, preprocessing, training, and evaluation.

The data acquisition phase begins with bridge instrumentation, where acceleration responses are collected through a network of accelerometers (A_1 to A_n) installed at various locations on the bridge structure. These sensors capture ambient excitation responses. The collected dataset is then systematically divided into training, validation, and test data subsets to ensure comprehensive model development and evaluation.

The preprocessing phase involves several important operations to prepare the raw data for effective analysis. Initially, the signal processing module performs detrending and denoising operations on both

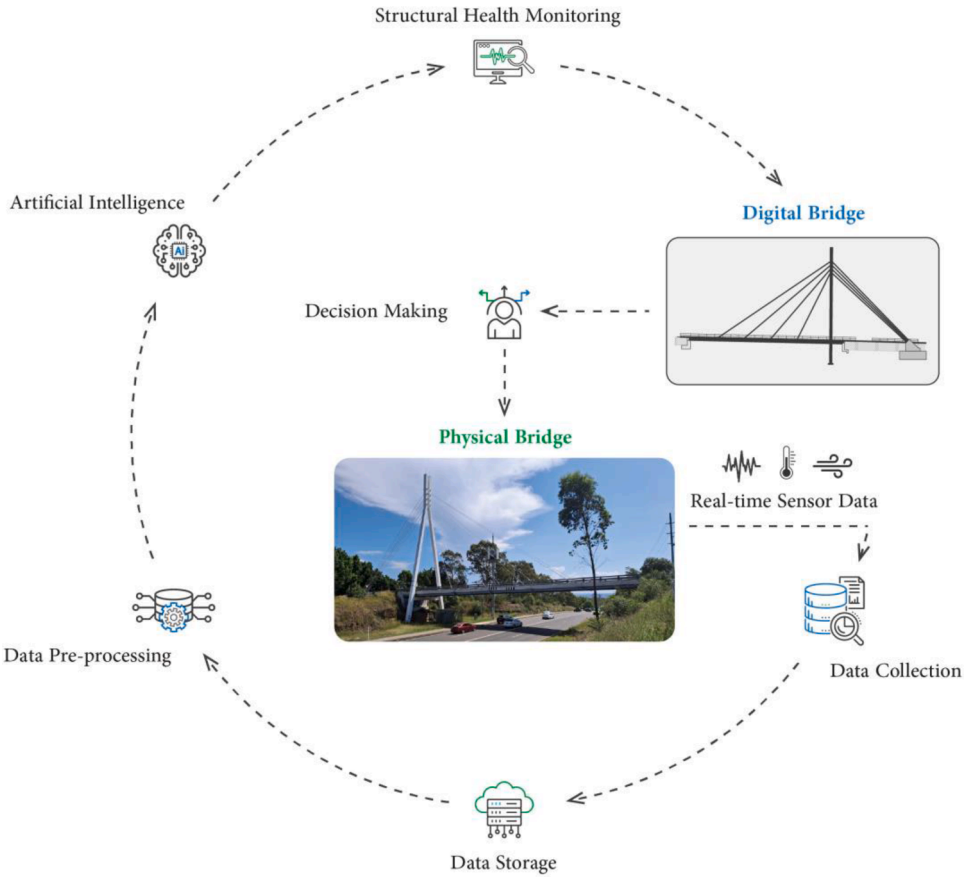


Fig. 1. The data-driven framework for AI-based bridge DT.

training and validation datasets to remove irrelevant noise and bias error in acceleration signals. Subsequently, feature extraction is performed across multiple domains, to capture complementary aspects of the signals. Lastly, the input configuration module prepares the data through scaling, windowing, batching, and structuring techniques to prepare the data for the model and training phase.

The training phase focuses on the core of the proposed methodology, the MDFA model. This specialized DL model integrates raw signals with extracted features from multiple domains to capture complex patterns within the data effectively. The MDFA comprises distinct encoders for each domain, which extract relevant features that are then fused together in a fusion model. Following feature fusion, dimensionality reduction techniques further refine these features, enabling the autoencoder to reconstruct the original inputs accurately. Training this model involves optimizing it to minimize reconstruction errors, enhancing its ability to detect anomalies through subtle deviations in signal patterns.

During the evaluation phase, synthetic data generation is performed using a transformer-based GAN model. The GAN model is trained using the test data, which is the same as the validation data. This innovative approach addresses data scarcity challenges common in SHM by preserving more real data for model training while still enabling thorough evaluation with unseen signals. The GAN generates two synthetic datasets: one representing normal structural conditions and another that served as a foundation for simulating various anomaly scenarios. This strategy is particularly valuable given the critical challenge in SHM regarding the lack of real-world datasets representing the damage state of civil infrastructure. The synthetic signals, which maintain the same characteristics as real data, undergo processing through the preprocessing algorithm, the same procedure applied in the preprocessing phase.

The trained MDFA model then reconstructs both normal and abnormal synthetic signals, and the reconstruction loss is calculated. A statistical assessment method evaluates the differences between original and reconstructed signals, determining the damage index for each sensor location. The outcome is visualized as an index across different sensor locations, identifying areas of potential structural damage

4.1. Data preparation

The dataset preparation process begins with the instrumentation of a bridge structure using an array of accelerometers (A_1 to A_n) positioned at strategic locations. These sensors continuously measure and record the acceleration responses of the bridge under ambient excitation. In this study, the measured acceleration signals were organized to facilitate the training and evaluation of the proposed methodology. The collected dataset was divided into two parts, where 70 % of the data was allocated for training and the remaining 30 % designated for validation and testing purposes. To maintain a large training set while ensuring thorough model evaluation, an innovative approach was adopted. Rather than further dividing the data into separate validation and test sets, which would reduce the available training data, the 30 % portion was utilized in two ways. This portion was employed as validation data in its original form, while also being used to train a GAN that produced synthetic samples for testing. This strategy enabled more real data to be preserved for training while sufficient data for both validation and testing phases was maintained.

4.2. Signal processing

Prior to feature extraction, the acceleration signals undergo two essential pre-processing steps, including detrending and low-pass

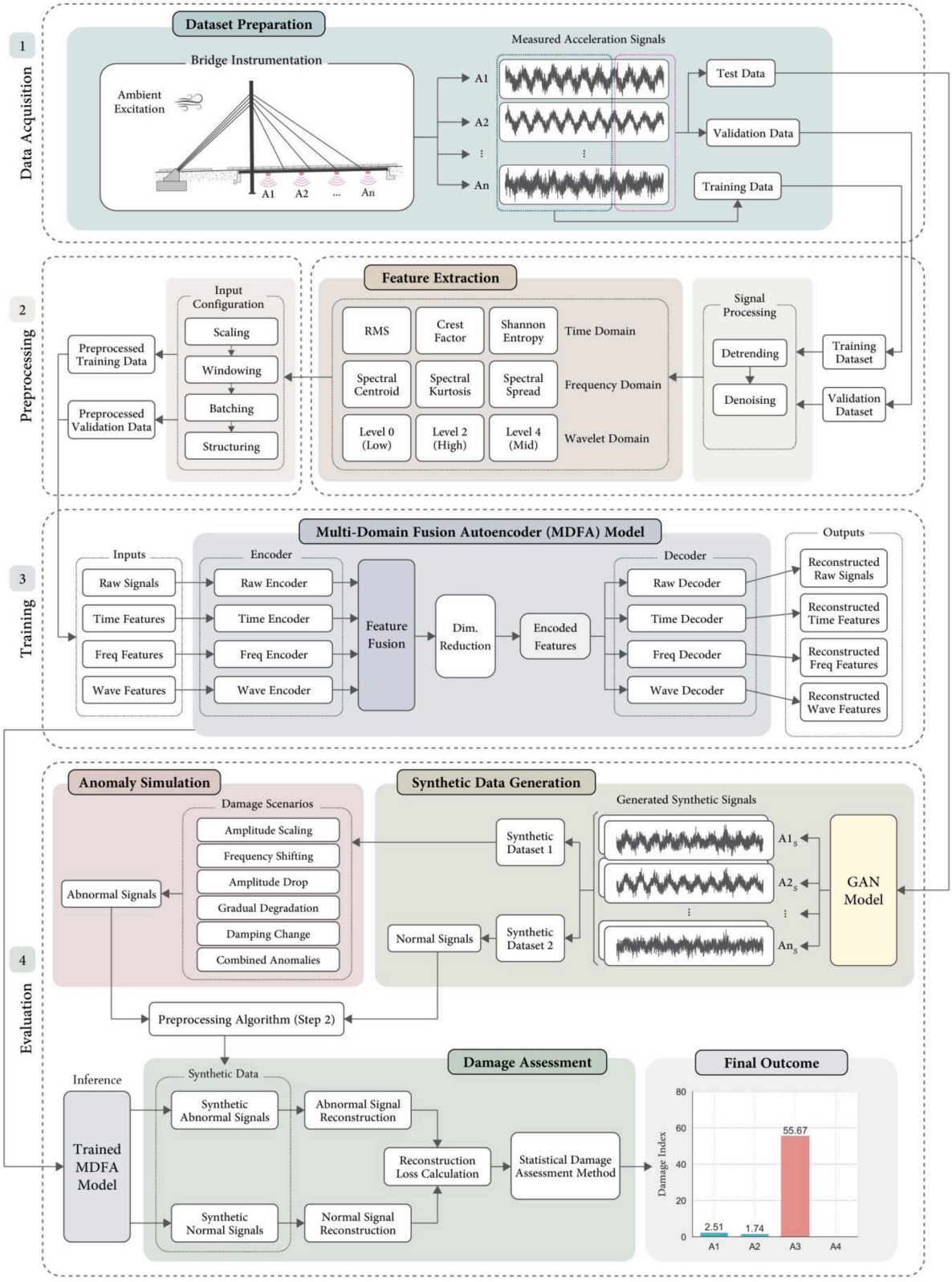


Fig. 2. The overall flowchart of the proposed methodology.

filtering. These steps are crucial for ensuring the quality and reliability of the extracted features.

Detrending is implemented to address inherent sensor errors that commonly occur in bridge monitoring applications. Accelerometers are susceptible to bias and drift errors due to environmental factors such as temperature fluctuations and humidity changes. While bias manifests as a constant offset from the true mean value, drift presents as a gradual, time-varying deviation in the measurements. Both errors negatively impact the performance of DL models by introducing artificial trends. To mitigate these effects, ordinary least squares regression (OLSR) [45] is

applied to identify and remove these errors from the acceleration signals, ensuring the subsequent analysis focuses on the structural response characteristics rather than sensor errors [25].

A low-pass filter is subsequently applied to the detrended signals to eliminate high-frequency noise components while preserving the meaningful structural response frequencies. The filter design incorporates a fourth-order Butterworth filter, chosen for its maximally flat frequency response in the passband [46]. The cut-off frequency is dynamically determined based on the signal's bandwidth, which is estimated using Welch's method for power spectral density estimation. The bandwidth calculation considers the frequency at which 90 % of the signal's power is contained, ensuring that the filter preserves the essential structural response components while effectively removing noise.

4.3. Feature extraction

Feature extraction is a critical step in the proposed methodology, as it transforms raw acceleration signals into meaningful representations that characterize the structural behaviour. To achieve this, features are extracted from three complementary domains, including time, frequency, and wavelet. This multi-domain approach ensures a comprehensive analysis by capturing both localized and global behavioural changes, which are essential for detecting subtle anomalies and structural damage and its location in bridge structures [47]. The process begins with a sliding window approach applied to the pre-processed signals. The window size and overlap are configurable to balance computational efficiency and temporal resolution. The features are computed for each window and subsequently interpolated to match the original signal length, providing continuous feature patterns.

In the time domain, features such as Root Mean Square (RMS), Crest Factor, and Shannon Entropy characterize the vibration signal's intensity, impulsiveness, and complexity, respectively. RMS quantifies the overall vibration energy, Crest Factor identifies transient peaks linked to localized damage, and Shannon Entropy reflects the randomness of the signal, capturing irregularities caused by structural changes. The frequency domain features, including Spectral Centroid, Spectral Spread, and Spectral Kurtosis, provide insights into the spectral distribution of the signal. These features track shifts in dominant frequencies, energy dispersion, and transient events, which are often indicative of changes in structural stiffness or localized damage. Lastly, wavelet-domain features are derived using a three-level decomposition with Daubechies-4 wavelets, enabling a multi-resolution analysis of the signal. Energy ratios at different levels capture the distribution of vibrational energy across high, mid, and low frequencies, effectively identifying localized and global anomalies at varying time scales. By combining features from these three domains, the proposed methodology creates a robust and diverse representation of the structural response.

The parameters utilized in the formulas are defined as follows, as detailed in Table 2. In the time domain, x_i represents the i-th sample of the signal, where $i = 1, 2, \ldots, N$ and N is the total number of samples in the signal. The RMS of the signal is denoted as x_{RMS}, which is calculated by taking the square root of the average of the squared values of the signal. The crest factor, CF, is defined as the ratio of the maximum absolute value of the signal to its RMS value. Shannon entropy, H, is computed based on the probability distribution p_j, which is derived from the signal histogram. In the frequency domain, the spectral centroid, C, represents the center of the mass of the power spectral density $P(f_k)$ at frequency f_k, where f_k is the frequency corresponding to the k-th spectral component. Spectral kurtosis, K_f, measures the sharpness of the frequency distribution and quantifies the dispersion of the frequency components around the centroid, calculated as the standard deviation of the frequency distribution. In the wavelet domain, the energy ratio ER_j is computed for each decomposition level j. The energy at level j, denoted as E_j, is the sum of the squared wavelet coefficients at that level, while the total energy, $\sum_k E_k$, is the sum of the energies across all decomposition levels.

Table 2
Mathematical formulas for feature extraction.

Domain	Feature	Mathematical Definition	Mathematical Formula
Time	Root Mean Square	Quantifies signal energy content	$x_{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
	Crest Factor	Ratio of peak amplitude to RMS value	$\mathrm{CF} = \frac{\max(\lvert x_i \rvert)}{x_{RMS}}$
	Shannon Entropy	Measure of signal complexity and randomness	$H = -\sum_j p_j \log_2(p_j)$
Frequency	Spectral Centroid	Centre of mass of the frequency spectrum	$C = \frac{\sum_{k=1}^{N} f_k P(f_k)}{\sum_{k=1}^{N} P(f_k)}$
	Spectral Kurtosis	Measure of sharpness in frequency distribution	$K_f = \frac{\sum_{k=1}^{N} (f_k - C)^4 P(f_k)}{\sigma_f^4 \sum_{k=1}^{N} P(f_k)}$
	Spectral Spread	Dispersion of frequency components around centroid	$\sigma_f = \sqrt{\frac{\sum_{k=1}^{N} (f_k - C)^2 P(f_k)}{\sum_{k=1}^{N} P(f_k)}}$
Wavelet	Wavelet Energy Ratios	Represents frequency variations	$ER_j = \frac{E_j}{\sum_k E_k}$

4.4. Multi-Domain fusion auto encoder (MDFA) model

The MDFA model provides a new multi-modal data integration approach to extract and fuse hidden representations from different sources. In this model, as shown in Fig.3, each data domain is processed through a dedicated encoder that transforms the input data into a domain-specific representation. This is an important step in order to preserve essential domain-specific features and capturing the intrinsic characteristics of input data.

Following the encoding stage, a fusion layer combines these individual latent representations into a shared latent space. The fusion process is designed to capture both general and specific aspects of different data domains. Attention mechanisms or constraint-based aggregation techniques are employed to ensure that the shared representation is both coherent and discriminative. The reconstruction phase involves a set of domain-specific decoders tasked with reconstructing the original inputs from the fused latent space. The MDFA model is trained by minimizing a composite loss function that balances reconstruction fidelity and fusion consistency.

4.4.1. Input configuration

The pre-processing phase transforms the input data into a structured format suitable for training the MDFA network. This phase consists of four main steps, including data scaling, sliding window segmentation, batch preparation and data structuring.

To normalize the data, a sensor-wise scaling method is used to handle potential outliers while preserving the relative relationships between features from each accelerometer. Sensor-wise scaling refers to applying normalization independently to each accelerometer's data rather than globally across all sensors. This approach scales the features based on robust statistical measures, ensuring that the influence of outliers is minimized. The normalization is performed independently for each sensor's features across all domains. The temporal relationships in the processed features are preserved through a fine-grained sliding window technique. Each feature sequence is further segmented into overlapping windows, allowing for comprehensive capture of temporal patterns while maintaining sufficient data resolution for training. This windowing process is applied uniformly across all feature domains, ensuring consistent temporal representation of the structural behaviour. To facilitate efficient model training, the pre-processed data is organized into a structured format using Tensor Flow's data pipeline. The data is

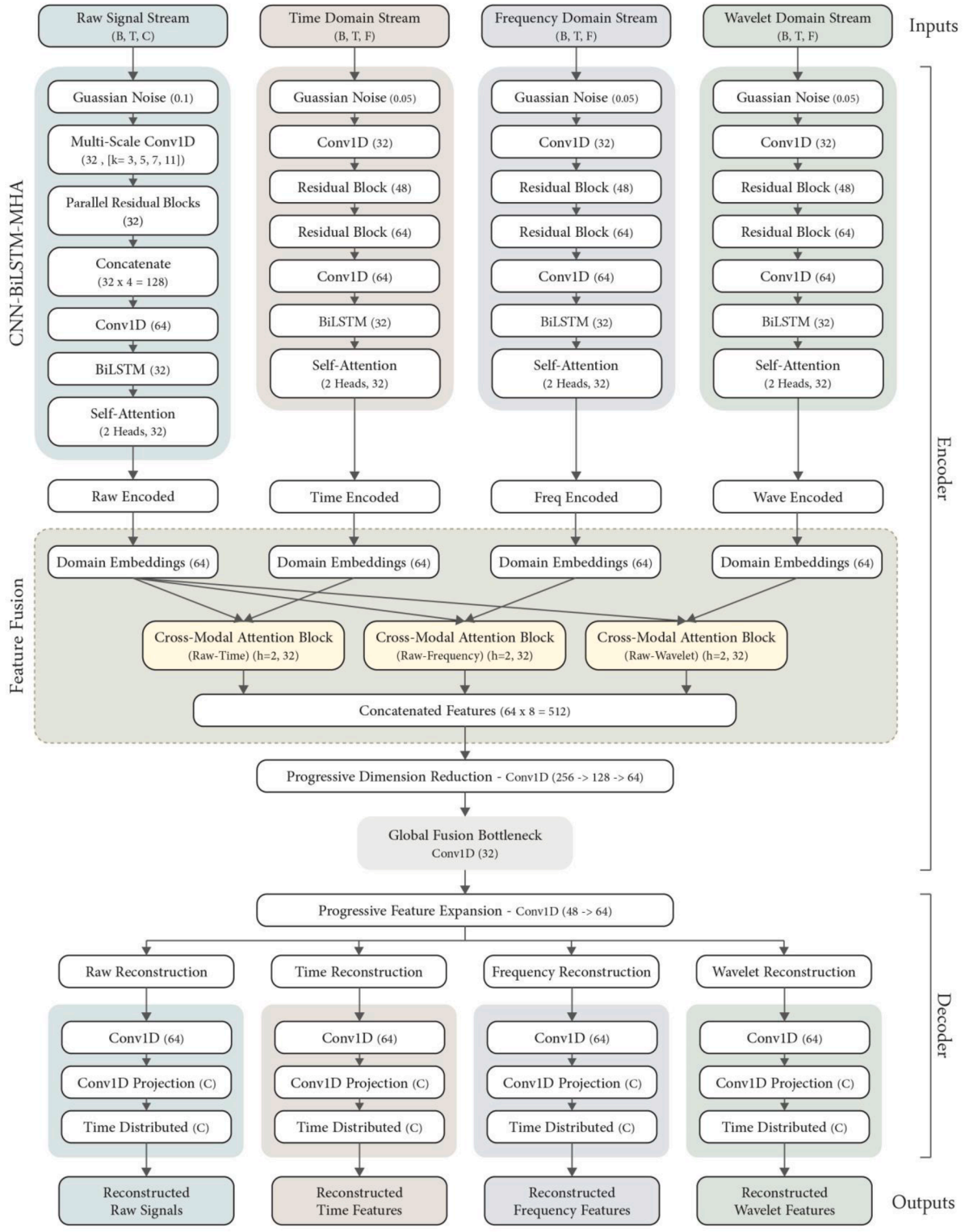


Fig. 3. The MDFA model Architecture.

batched with a size of 512 samples and configured with prefetching to optimize training performance. Subsequently, the data structuring step organizes the processed data into corresponding input and output dictionaries for each accelerometer and feature domain (raw, time, frequency, and wavelet), creating the multimodal input configuration required by the MDFA network architecture. By following these preprocessing steps, the data was transformed into a format that is both normalized and structured, making it suitable for direct input into the model for training or prediction.

4.4.2. CNN-BiLSTM-MHA model architecture

The proposed model architecture employs a novel multi-stream hybrid approach that processes both raw acceleration signals and their domain-transformed representations simultaneously. While raw acceleration data contains the original structural response patterns, domain-transformed features (time, frequency, and wavelet domains) offer

complementary perspectives of the signal characteristics. The simultaneous processing of these complementary features enables the model to capture a more comprehensive representation of the structural health state. The architecture consists of three main components, including an encoder with multi-stream processing, a feature fusion mechanism, and a decoder for multi-domain reconstruction, as illustrated in Fig. 3. In this Figure, (B, T, C) refers to batch size, time step, and channel, respectively, while (B, T, F) also represents batch size, time step, and features, respectively and the dimensions of each layer are explicitly shown.

Each stream implements a CNN-BiLSTM-MHA (Convolutional Neural Network-Bidirectional Long Short-Term Memory-Multi-Head Attention) hybrid model [48]. To address the inherent uncertainties in real-world structural data, Gaussian noise layer has been added during the training process, which serves as a regularization technique and significantly enhances the generalization capabilities of the model. The raw signal stream processes the original acceleration data through a multi-scale Conv1D layer with varying kernel sizes (3, 5, 7, 11) to capture temporal patterns at different scales, while all other Conv1D operations throughout the streams utilize a kernel size of 3. After each convolution, Layer Normalization and an ELU activation are applied to stabilize training dynamics and introduce nonlinearity.

4.4.3. Feature fusion

This research focuses on integrating and collectively analysing features obtained from different domains using DL techniques to enhance real-time damage detection through adaptive, unsupervised learning on real-world bridge data. Instead of relying on only raw sensor data, combining meaningful information from data and the raw data itself provides a more comprehensive understanding of the structure behaviour. In this study, the feature fusion was conducted after feature extraction from signals.

A crucial component of the architecture is the Feature Fusion mechanism, which employs cross-modal attention blocks to intelligently combine information from different domains. The cross-modal attention operates by learning relationships between domain pairs (Raw-Time, Raw-Frequency, Raw-Wavelet), enabling the model to identify correlations and dependencies across different signal representations. This fusion approach allows the model to: (1) Learn complex relationships between features across different domains, (2) Automatically weight the importance of different domain representations, (3) Capture complementary information that might be missed when domains are processed independently and (4) Enhance the model's sensitivity to subtle anomalies that manifest across multiple domains

The global fusion bottleneck serves as a crucial integration point where features from all accelerometers are combined and compressed. The process begins by concatenating features from all accelerometers. This concatenated feature set undergoes progressive dimension reduction through two 1D convolution layers with kernel sizes of 1, first reducing to 128 filters and then to the final bottleneck dimension of 32 filters. This bottleneck design ensures that the model learns to preserve the most essential characteristics of the structural response while establishing relationships between different sensor locations, enabling the detection of global structural behaviour patterns and anomalies.

4.4.4. Decoder

The decoder architecture implements a reconstruction pathway that progressively expands the compressed features back to the original domain dimensions. The decoder is a branched architecture, creating separate reconstruction pathways for each accelerometer and domain combination. Each pathway concludes with a domain-specific projection layer that maps the features to the appropriate output dimensions through a time distributed dense layer. The time distribution layer in the final output ensures proper temporal alignment of the reconstructed signals across all domains and accelerometers. This comprehensive reconstruction approach enables the model to validate its learned representations and provide multiple perspectives for anomaly detection through reconstruction error analysis.

5. Evaluation strategy

5.1. Synthetic data generation

To overcome the limitation of insufficient real-world data in bridge health monitoring for training and evaluation of unsupervised DL models, a transformer-based time-series Wasserstein generative adversarial network with gradient penalty (TTS-WGAN-GP) was employed [49]. This model demonstrated its capacity to generate realistic synthetic acceleration signals representative of the bridge's structural responses.

The synthetic data generation process was structured to create two distinct datasets. The first dataset was generated to represent the normal operating conditions of the bridge, capturing the typical dynamic responses under ambient excitation. The second dataset served as a foundation for simulating various anomaly scenarios, providing a controlled environment for evaluating the proposed methodology's ability to detect structural anomalies.

5.1.1. TTS-WGAN-GP architecture

The TTS-WGAN-GP architecture, illustrated in Fig. 4, consists of two primary components: a generator and a critic, both leveraging transformer encoder blocks to effectively capture temporal dependencies in acceleration time-series data.

The generator component begins with a random noise vector input (z) which undergoes latent expansion to create a foundation for the synthetic signal. This expanded representation then receives positional encoding, which injects sequential information that helps the model understand temporal relationships within the data. The core of the generator consists of multiple transformer encoder blocks arranged in sequence. Each block contains layer normalization, multi-head attention mechanisms, and feed-forward neural networks with residual connections. These components work in concert to transform the noise input into meaningful temporal patterns that mimic real acceleration signals. The final output produces synthetic acceleration signals. The Critic component evaluates both real acceleration signals and the generator's synthetic outputs. After an initial input projection layer that maps the signals to a higher-dimensional space, the data flows through parallel transformer encoder blocks. Similar to the generator, these blocks contain layer normalization, multi-head attention, and feed-forward components with residual connections. The transformer blocks enable the critic to analyse the data at multiple temporal scales, capturing both local vibration patterns and global structural response characteristics. The processed features are then condensed through an attention pooling mechanism, which produces a context-aware representation of the input signals.

Finally, a Wasserstein score is computed, quantifying the distributional distance between the real and synthetic data distributions. The model's configuration included several key hyper-parameters: a sequence length of 600 time steps, a latent dimension of 128, and a hidden dimension of 256 for both generator and critic networks. The generator utilized 4 transformer layers with 4 attention heads, while the critic employed 3 transformer layers with 8 attention heads, allowing it to evaluate signals from multiple perspectives. A consistent dropout rate of 0.1 was applied throughout the network to prevent overfitting. During training, the generator used a learning rate of $5e^{-5}$ while the critic used $2e^{-4}$, with a gradient penalty coefficient of 10 to ensure training stability. The critic was updated 5 times for each generator update, following standard practices for WGAN training.

The TTS-WGAN-GP model is trained on real-world validation data, enabling it to inherently learn and replicate the uncertainties and variabilities present in these actual signals. Thus, the generated synthetic data naturally embodies these real-world characteristics, with the GAN's noise input facilitating diverse sampling from this learned distribution.

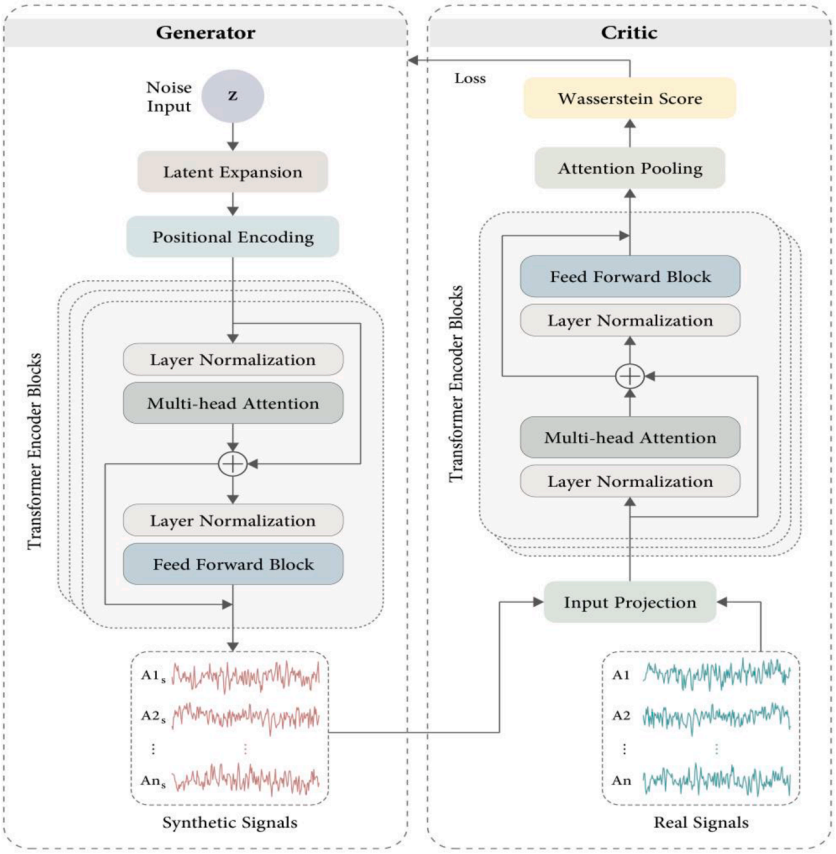


Fig. 4. The TTS-WGAN-GP Architecture.

5.1.2. TTS-WGAN-GP data validation

To ensure the quality and reliability of the synthetically generated acceleration signals, a comprehensive validation approach was employed using multiple similarity metrics across both time and frequency domains. This validation process was crucial to verify that the synthetic data accurately represented the statistical and physical characteristics of real bridge vibration responses [50].

In the time domain, two primary metrics were utilized to evaluate the similarity between real and synthetic signals. The Fréchet Inception Distance (FID) was adopted to quantify the statistical distance between the distributions of real and synthetic data. This metric calculates the differences between the means and standard deviations of the feature representations, providing a scalar value where lower scores indicate greater similarity. Additionally, Average Cosine Similarity was implemented to measure the directional similarity between feature vectors extracted from real and synthetic signals. This approach involved extracting statistical features (including median, mean, standard deviation, variance, root mean square, maximum, and minimum values) from both datasets and computing the normalized dot product between these feature vectors.

For frequency domain validation, two complementary methods were employed to ensure that the synthetic signals accurately captured the spectral characteristics of bridge vibrations. Power Spectral Density (PSD) Cosine Similarity compared the frequency content of the signals by calculating the cosine similarity between the power spectral density estimates of real and synthetic data. This metric specifically evaluates how well the synthetic signals replicate the energy distribution across different frequencies, which is particularly important for structural vibration analysis. The second frequency domain metric, Magnitude Correlation, assessed the linear correlation between the magnitude spectra of the Fourier transforms from both datasets, providing insight into how well the synthetic signals preserved the dominant frequency components characteristic of the bridge's dynamic behaviour.

These four validation metrics collectively provided a robust framework for assessing the TTS-WGAN-GP model's performance in generating realistic bridge acceleration signals. By validating similarity in both time and frequency domains, the approach ensured that the synthetic data captured not only the statistical properties of the signals but also the underlying physical dynamics of the structural system.

5.2. Synthetic anomaly simulation

SHM primarily relies on accurately identifying changes in structural behaviour that indicate potential damage or deterioration. A critical challenge in SHM lies in the lack of real-world datasets representing the damage state of civil infrastructure. This scarcity of damaged-state data creates a fundamental obstacle for developing robust anomaly detection algorithms, particularly when employing unsupervised DL approaches. To overcome this limitation, a comprehensive framework is proposed for simulating various structural anomalies using acceleration data obtained from a real-world bridge under ambient excitation. The proposed method enables systematic evaluation of anomaly detection models by introducing controlled modifications to the structural response characteristics. These modifications are designed to replicate specific anomaly patterns commonly associated with different types of structural damage [51–53]. By creating a diverse set of simulated anomaly scenarios with varying characteristics, we can evaluate the sensitivity and accuracy of detection algorithms across different damage modalities. The proposed methodology enables systematic evaluation of anomaly detection models by introducing controlled modifications to the structural response characteristics. These modifications are designed to replicate specific anomaly patterns commonly associated with different types of structural damage. By creating a diverse set of simulated anomaly scenarios with varying characteristics, the sensitivity and accuracy of

detection algorithms across different damage modalities can be evaluated. Table 3 presents the mathematical formulations for each implemented anomaly scenario along with brief descriptions of the physical phenomena they represent.

The parameters used in the mathematical formulations presented in Table 3 are defined as follows. The original and modified acceleration signals in the time domain are denoted as $x(t)$ and $\tilde{x}(t)$, respectively, while their frequency-domain counterparts are $x(f)$ and $\tilde{x}(f)$. The Fourier transform and its inverse are represented by $\mathscr{F}$ and $\mathscr{F}^{-1}$. Amplitude-related modifications are controlled by the scaling factor S_f and the drop percentage D_p, which determine amplitude amplification and localized reductions, respectively. The frequency shift parameter f_s alters the modal properties, while damping changes are characterized by δ, modifying spectral amplitude near the dominant frequency. The mean value of the signal segment is given by μ, serving as a reference for amplitude drop and gradual degradation. In the combined anomaly scenario, transformations are represented by $\mathscr{T}_D$, $\mathscr{T}_F$, and $\mathscr{T}_A$, corresponding to damping change, frequency shift, and amplitude scaling, respectively.

Following the generation of a set of simulated anomaly scenarios with varying characteristics, the corresponding synthetic acceleration signals undergo two essential processing stages: pre-processing and feature extraction. In the pre-processing stage, the signals are first detrended and then subjected to low-pass filtering, as detailed in Section 4.2. Subsequently, feature extraction is performed to transforms anomaly simulation signals into meaningful representations. Therefore, features are extracted across three complementary domains including time, frequency, and wavelet, as described in Section 4.3.

5.3. Damage evaluation metric

After training of the MDFA model, the model is then employed to reconstruct the signals, and the reconstruction error (E) is calculated using the Mean Squared Error (MSE) of the input input (x) and reconstructed ($\hat{x}$) signals, as follows:

$$E = (x - \hat{x}) \tag{1}$$

This error typically follows a Gaussian distribution, with the Probability Density Function (PDF) expressed as follows:

$$f_x(E) = \frac{1}{\sigma\sqrt{2\pi}}\ exp\left(-\frac{(E-\mu)^2}{2\sigma^2}\right) \tag{2}$$

where σ and μ are the mean and standard deviation of E, respectively. To evaluate the amount of information represented by the PDF computed using Eq. (2), a measure of entropy, specifically Rényi entropy, is commonly used [54]. This entropy measure helps quantify the level of uncertainty within the set of descriptors. The definition of Rényi entropy is outlined below.

Table 3
Simulated anomaly scenarios and their specifications.

Anomaly Scenario	Mathematical Formulation	Simulated Effect
Amplitude Scaling	$\tilde{x}(t) = (1 + 0.5\ S_f)\cdot x(t)$	Stiffness reduction or mass changes
Frequency Shifting	$\tilde{x}(t) = \mathscr{F}^{-1}\{\mathscr{F}\{x(t)\}\cdot e^{-j2\pi f_s t}\}$	Changes in natural frequencies or modal shift
Amplitude Drop	$\tilde{x}(t) = \mu + [x(t) - \mu]\cdot(1 - D_p)$	Localized reductions or abrupt changes
Gradual Degradation	$\tilde{x}(t) = \mu + [x(t) - \mu]\cdot x(t)$	Progressive deterioration such as fatigue or corrosion
Damping Change	$\tilde{x}(f) = x(f)\cdot\delta$	Damping variations
Combined Anomalies	$\tilde{x}(t) = \mathscr{T}_D(\mathscr{T}_F(\mathscr{T}_A(x(t))))$	Complex damage scenarios

Assuming X as a random variable with a constant distribution of $f_x(x)$, the Rényi entropy of order α (where $\alpha \geq 0,\ \alpha \neq 1$ and $\alpha \neq 0$) can be calculated as follows:

$$Renyi(X) = \frac{1}{1-\alpha} log\left(\int (f_X(x))^{\alpha} dx\right) \tag{3}$$

The entropy of a PDF can be interpreted as a measure of both uncertainty and information content. As α approaches infinity, the Rényi entropy is dominated by the events with the highest probabilities. The Rényi entropy of order α places greater emphasis on less probable data, making it a useful tool to measure the information content in each record. Thus, it can be utilized to distinguish between special differences and normal ones in our damage detection task.

By substituting Eq. (3) into Eq. (2), the Rényi entropy with $\alpha = 2$ could be calculated by:

$$Reiny(E) = -log\left[\int \left(\frac{1}{\sigma\sqrt{2\pi}}\ exp\left(-\frac{(E-\mu)^2}{2\sigma^2}\right)\right)^2 dx\right] \tag{4}$$

It is important to note that the analytical form of Rényi entropy is not derived in Eq (4). Instead, a non-parametric formulation of Rényi entropy is used to evaluate the full error distribution in the latent space. The applied Rényi entropy is more efficient in comparison to widely used Damage-Sensitive Features (DSF) applied in previous studies [25,55], as it considers the full error distribution rather than only the peak and variance, allowing detection of subtle changes (even minor damage) that might be overlooked by Standard deviation to peak PDF ratio (SDPP) method [55]. Rényi entropy captures higher-order statistical characteristics of the error distribution and enables the detection of subtle yet systematic deviations in signal reconstruction error. These variations are often overlooked by conventional metrics and affect the sensitivity of early-stage anomaly detection. In comparison to traditional loss functions, such as standard MSE, Kullback–Leibler (KL) divergence [56], and the SDPP, Rényi entropy provides a more holistic representation of the underlying distribution [57]. MSE is limited to pointwise amplitude differences, while KL divergence relies on a well-defined reference distribution and is particularly sensitive to regions of low probability. SDPP, on the other hand, reduces the distribution to a single moment ratio, thereby losing significant distributional details. By contrast, Rényi entropy provides a robust statistical measure capable of capturing complex distributional shifts and its sensitivity in detection of subtle variations makes it particularly effective for identifying early-stage or small-scale damages. The threshold for anomaly detection was determined using a percentile-based approach applied to Rényi entropy values calculated from the normal data. After computing the Rényi entropy for each accelerometer under normal operating conditions, a statistical threshold was established at a high percentile (typically the 99th percentile) of this distribution. This methodology ensures that values exceeding the threshold represent statistically significant deviations from the normal pattern.

6. Case Study I: Werrington Bridge, NSW Australia

6.1. Dataset

For the first case study, the acceleration data from the Werrington bridge have been utilized. This bridge, as shown in Figure 5(a), crosses the Great Western Highway and connects the north and south Penrith campuses of Western Sydney University in Sydney. This Bridge is a cable-stayed structure with a 46.2 m span, with a structural system consists of a composite steel-concrete deck supported by an A-shaped steel pylon. The deck, with a uniform width of 6.3 m, is supported by four longitudinal girders, which are internally connected by seven transverse cross girders. Additionally, steel corrugated sheets are embedded within the deck to enhance its composite structural performance. The deck is also supported by three lines of four bearings and is

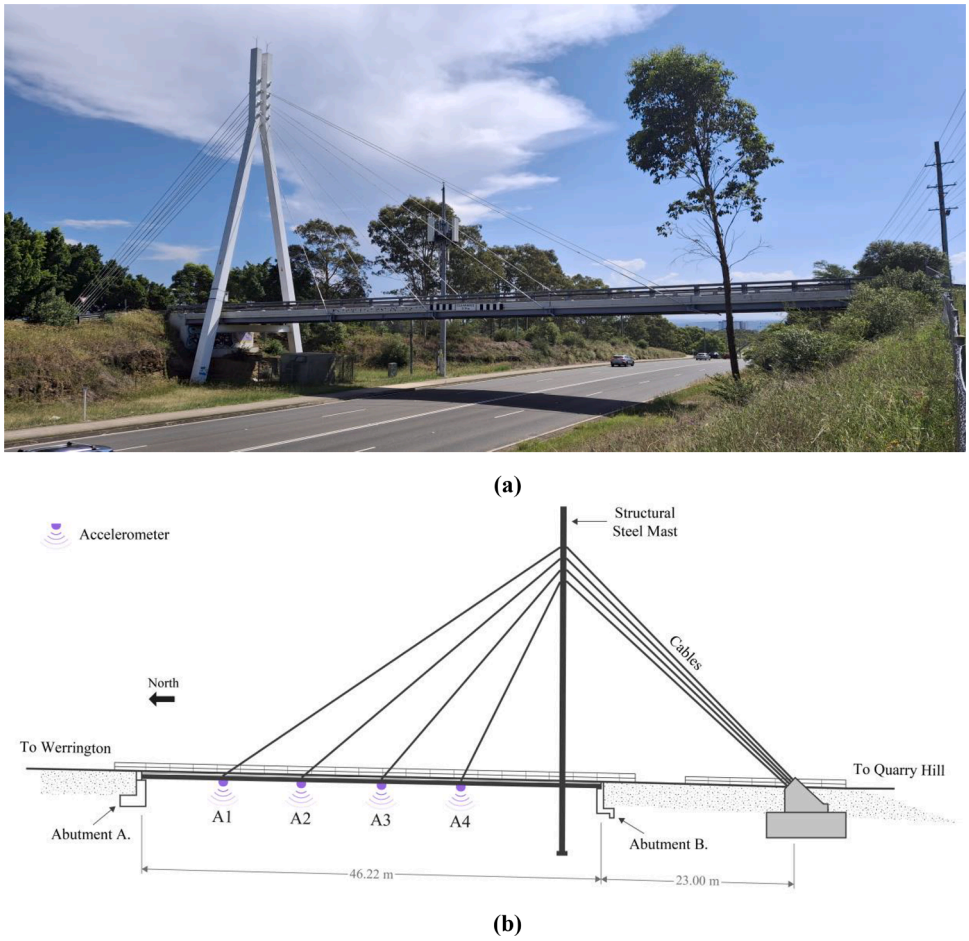


Fig. 5. (a): Werrington bridge, (b) Schematic illustration of sensor placement.

stabilized using two sets of eight stay cables: one set anchored to the upper end of the pylon and another to a concrete block at the southern end.

To monitor the bridge's structural vibrations, a network of high-precision accelerometers was installed at different locations across the bridge, as depicted in Figure 5(b). These sensors consistently recorded acceleration data, reflecting the vibrational characteristics of the structure. The detailed descriptions of the instrumentation and data collection procedures are provided in [58]. In this case study, 5 minutes of vibration measurements were used for training, and 2 minutes for validation and testing.

6.2. Extracted Features from Bridge Acceleration Data

The acceleration signals obtained from the bridge monitoring system required pre-processing to ensure data quality before feature extraction. As described in Section 4.2, the raw signals exhibited bias error, which was eliminated using OLSR method. Following detrending, a low-pass Butterworth filter was applied to remove high-frequency noise components while preserving the essential structural response characteristics. The selection of an appropriate cut-off frequency for the low-pass filter was determined through bandwidth estimation using the Welch method. The bandwidth analysis revealed that at least 90% of the signal power for all accelerometers was concentrated below 200 Hz, which was subsequently chosen as the cut-off frequency. This filtering approach effectively preserved the meaningful structural frequencies while attenuating noise components. It is important to note that this filtering process was specifically applied to the feature extraction procedure and did not affect the detrended raw signals, which retained their original frequency content for damage detection analysis. This approach avoids losing any potentially valuable information and prevents overfitting during model training by allowing the raw data to retain its full dynamic range.

Figure 6 presents the pre-processed acceleration signals and their corresponding extracted features across multiple domains. All signals and features were scaled for better visualization and comparison. The top panel shows the detrended raw acceleration measurements from all four sensors (A1-A4), revealing bridge's dynamic response patterns. The subsequent panels illustrate the extracted features in time, frequency, and wavelet domains for each sensor, computed according to the equations presented in Table 3. The extraction process employed a sliding window approach with a window size of 300 samples and 50% overlap, striking a balance between computational efficiency and temporal resolution. These extracted features formed the foundation for training the MDFA model, providing a comprehensive representation of bridge's dynamic behaviour under normal operating conditions.

6.3. MDFA Model Training Performance

The MDFA model was trained using an unsupervised learning approach with real-world acceleration data collected from the Werrington Bridge under normal operating conditions. This approach allowed the model to learn the typical patterns of a healthy structural state without requiring data from damaged conditions.

The training process began with thorough data preparation, as described in Section 4.4.1. Sensor-wise scaling was applied to normalize the data from each accelerometer independently while preserving the relationships among features. The pre-processed data was then

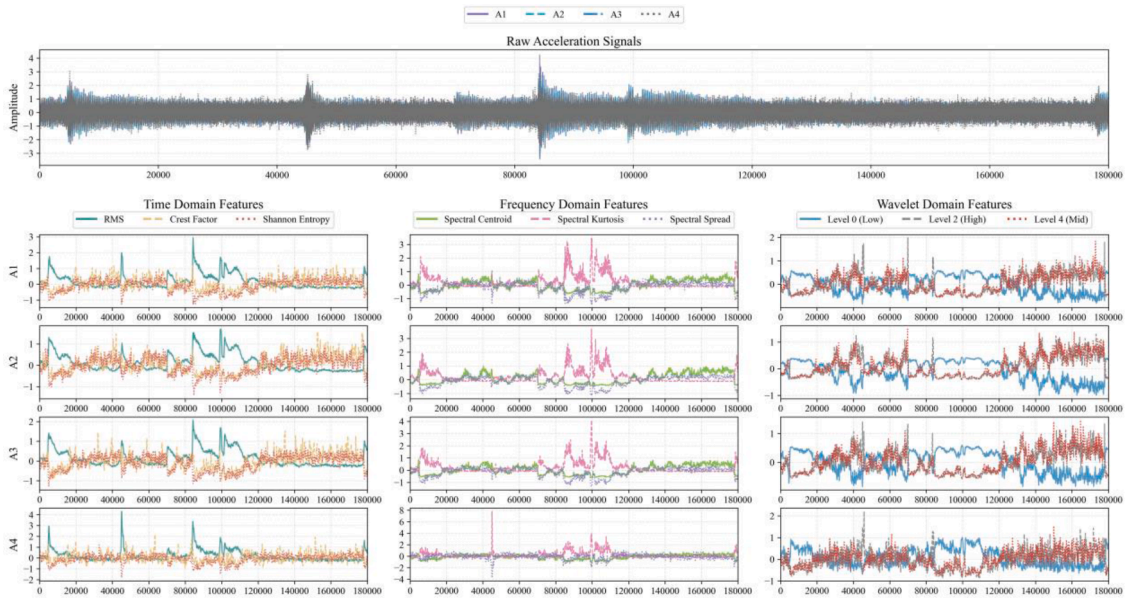


Fig. 6. Pre-processed acceleration signals from the Werrington Bridge and their extracted features across time, frequency, and wavelet domains.

organized using a sliding window of size 10 and grouped into batches of 512 samples via Tensor Flow's data pipeline. The input structure was configured to accept raw acceleration data with dimensions (512, 10, 1) and domain-specific features with dimensions (512, 10, 3), ensuring consistency in data format.

The model was trained by minimizing the mean squared error between the input signals and their reconstructions across all domains. A domain-specific loss weighting strategy was applied by assigning a weight of 1.0 to the reconstruction of raw acceleration signals and a weight of 0.8 to the reconstruction of extracted features (time, frequency, and wavelet). This scheme prioritized the accurate reconstruction of raw signals while incorporating the complementary information provided by the derived features. Optimization was performed using the Adam optimizer, which was initialized with a learning rate of 0.0001. Training was conducted for 100 epochs. To prevent overfitting and promote model generalization, early stopping was implemented based on the monitoring of the validation loss. In addition, a learning rate reduction strategy was implemented to decrease the learning rate dynamically when the validation loss plateaued.

Figure 7 shows the overall training and validation loss curves over 100 epochs. Both loss curves decreased steadily, which suggests that the model effectively learned to reconstruct the input data. The convergence of the training and validation losses implies that the model generalized well to unseen data from normal operating conditions. The overall training loss, representing the aggregate of all accelerometer losses across all domains, was 0.0839, while the validation loss was 0.1516. Training was executed on a PC equipped with an RTX2080 GPU and 128 GB of RAM, requiring a total time of 7 hours and 12 minutes.

Figure 8 also displays the loss curves for each accelerometer (A1, A2, A3, and A4) and for each domain (raw, time, frequency, and wavelet). These detailed curves offer a granular view of the model's performance. Notably, the validation loss for accelerometer A4 was consistently higher than its training loss and those of the other accelerometers. This discrepancy may be attributed to the elevated noise level in A4's acceleration data. Specifically, the signal-to-noise ratio for A4 was calculated at 3.8 dB, which is significantly lower than the approximately 8 dB observed for the other accelerometers. The increased noise in A4's data posed a greater challenge for the reconstruction process, resulting in a higher validation loss. Nonetheless, the model demonstrated its ability to learn the underlying patterns, as reflected by the overall decrease in the A4 loss curves. In summary, the MDFA model exhibited robust training performance and successfully captured the normal behavioural patterns of the bridge.

To further validate the reconstruction capabilities of the MDFA model, Figure 9 compares the actual and reconstructed signals for accelerometer A3 across all domains. The top panel displays the reconstruction of the raw acceleration signal, illustrating that the model may capture both the overall signal pattern and transient events accurately. Subsequent panels compare the reconstructed features in the time, frequency, and wavelet domains. The close alignment between the real and reconstructed signals across all domains indicates that the model successfully captured the underlying patterns associated with normal operational behaviour. This high-fidelity reconstruction establishes a reliable baseline for damage detection analysis, as significant deviations from these reconstruction patterns may indicate potential structural anomalies.

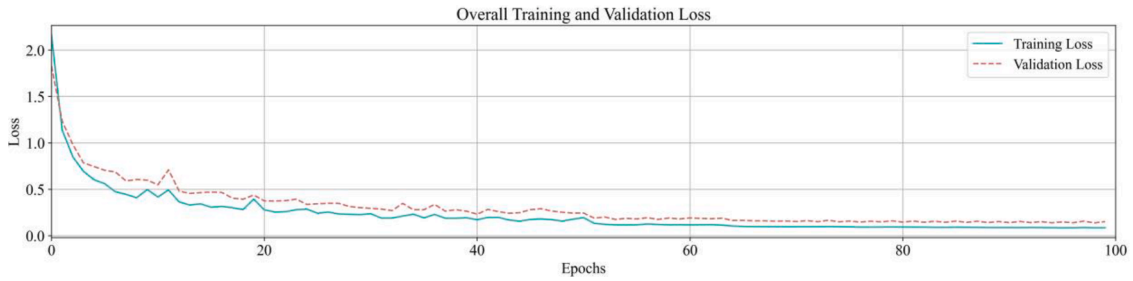


Fig. 7. Overall training and validation loss curves over 100 epochs.

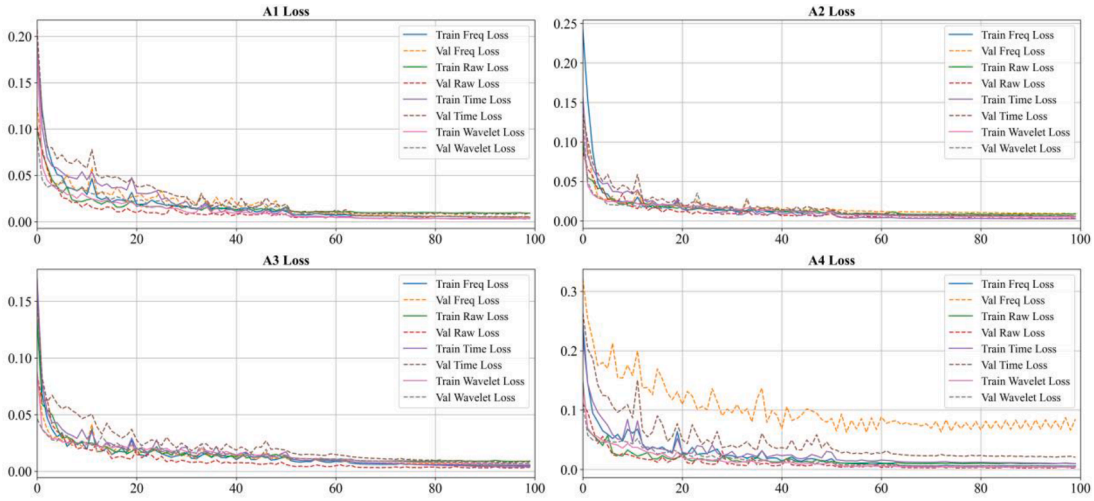


Fig. 8. Loss curves for A1, A2, A3, and A4 accelerometers over 100 epochs.

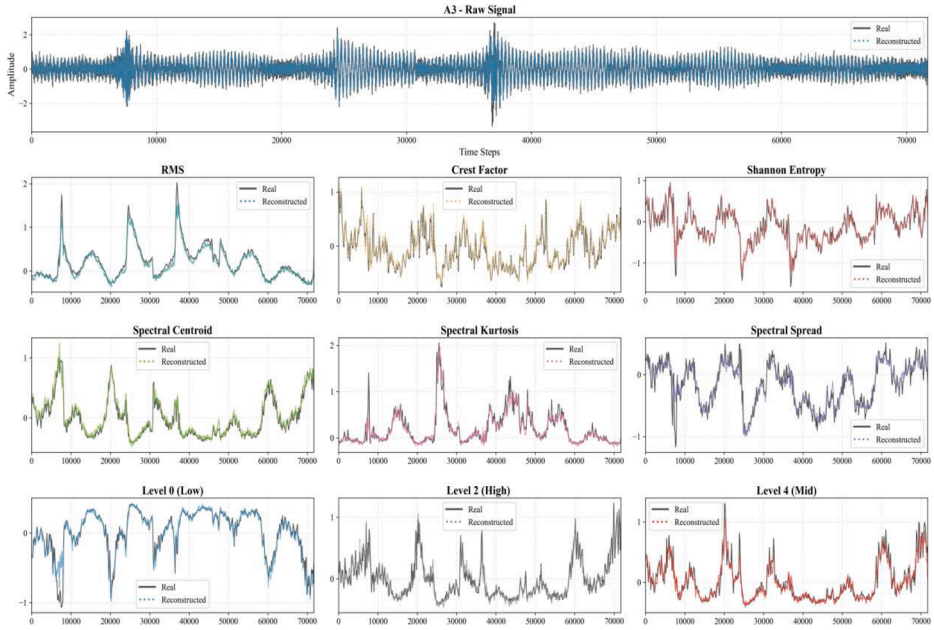


Fig. 9. Comparison of the actual and reconstructed signals for accelerometer A3.

6.4. Similarity Analysis of Synthetic and Real Acceleration Signals

The quality assessment of synthetic data is essential for validating the subsequent damage detection analysis. As detailed in Section 5.1, a TTS-WGAN-GP was employed to generate two datasets, one reflecting normal operating conditions and the other simulating damage scenarios. This section provides a comprehensive evaluation of synthetic data quality through both visual comparisons and quantitative metrics.

Figure 10 compares randomly selected windows of real and synthetic acceleration signals for all accelerometers in the time and frequency domains. The time-domain plots indicate that the synthetic signals capture the amplitude variations and temporal patterns characteristic of the real measurements. In the frequency domain, the synthetic signals accurately reproduce the bridge's primary natural frequency at approximately 2.3 Hz while also reflecting the broader spectral features observed in the real data.

Quantitative assessments were conducted using multiple metrics, as defined in Section 5.1.2, and the results are summarized in Table 4. The metrics reveal a high degree of similarity between the real and synthetic signals. FID values remained consistently low, and average cosine similarity values in the time domain were around 0.985. Similarly, in the frequency domain, both PSD cosine similarity and magnitude correlation values approached one. These findings corroborate the visual comparisons from Figure 10 and validate the realism of the synthetic data generated by the TTS-WGAN-GP model. The high fidelity of the synthetic signals provides a reliable basis for subsequent damage scenario simulations and for evaluating the MDFA model's damage detection capabilities.

6.5. Anomaly Simulation Results

The anomaly detection capabilities of the MDFA framework were evaluated by introducing a series of synthetic anomalies into the normal-condition data generated for the A3 accelerometer signal. Six distinct anomaly types were simulated at three severity levels, thereby capturing a range of structural changes that affect amplitude, frequency, and damping characteristics. These anomalies, detailed in Section 5.2, were systematically embedded in the synthetic acceleration signals produced

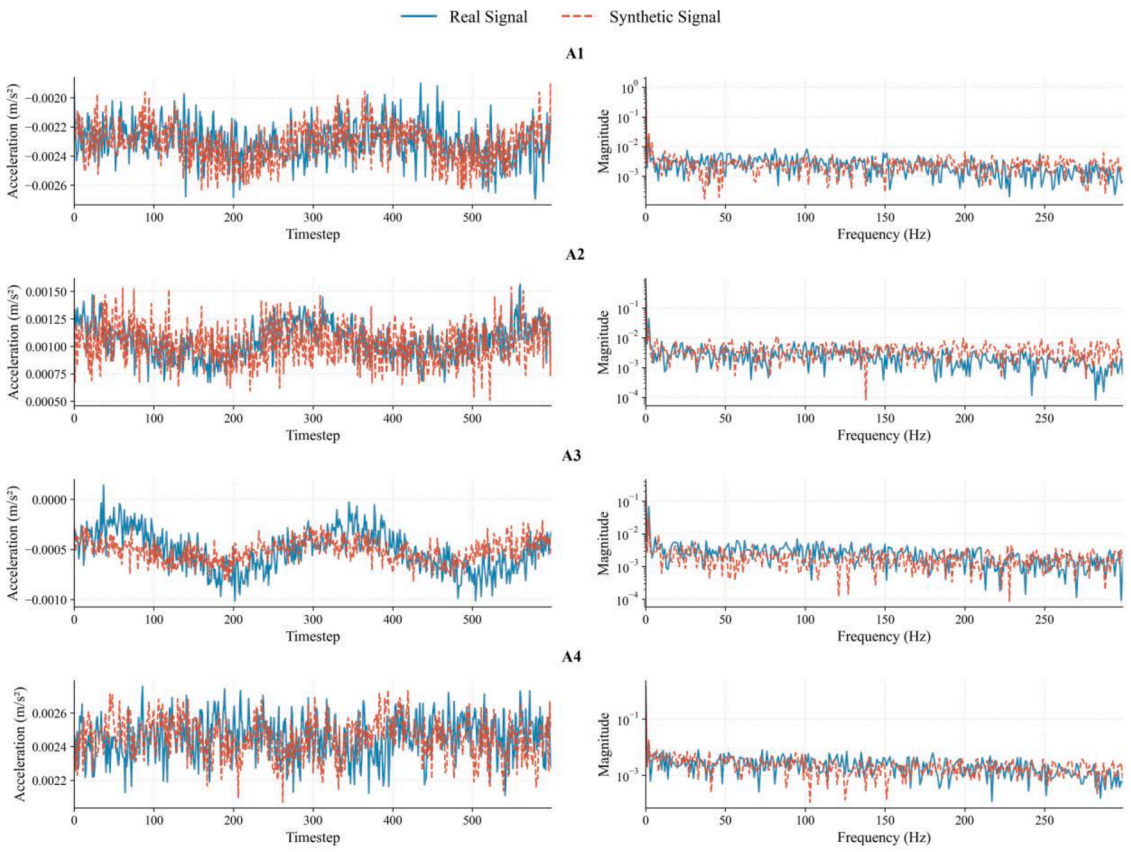


Fig. 10. Comparison of the randomly selected windows of real and synthetic signals.

Table 4
Similarity assessments results of real and synthetic signals using different metrics.

Accelerometer	Time Domain		Frequency Domain	
	FID	Avg Cosine Sim	PSD Cosine	Mag. Corr
A1	1.0293e-09	0.987027	0.971905	0.998821
A2	4.1435e-10	0.980625	0.948335	0.988910
A3	1.8947e-09	0.986023	0.966115	0.976641
A4	1.4895e-10	0.989055	0.937398	0.999400

by the TTS-WGAN-GP model. Table 5 provides an overview of the simulated scenarios and their corresponding characteristics.

As shown in Figure 11, the simulated anomalies manifest differently in the acceleration signals. Some changes are more visually apparent than others. The amplitude scaling and amplitude drop scenarios exhibit the most readily observable modifications. In contrast, frequency shifts and damping changes introduce subtle changes that are challenging to detect through visual inspection alone.

For frequency shifting and damping change scenarios, the relationship between the numerical magnitude of the change and its severity is inverse. This relationship follows a principle where the spatial concentration of the change often carries more significance than its magnitude.

Table 5
Overview of the simulated scenarios and their corresponding characteristics.

Anomaly Type	Severity	Magnitude of Change	Description
Amplitude Scaling	Light	+10 % increase	Uniform scaling of acceleration amplitude
	Moderate	+15 % increase	
	Severe	+20 % increase	
Frequency Shifting	Light	0.2 Hz shift	Changes in natural frequencies or modal shift
	Moderate	0.15 Hz shift	
	Severe	0.1 Hz shift	
Amplitude Drop	Light	15 % reduction	Sudden localized amplitude reductions
	Moderate	20 % reduction	
	Severe	25 % reduction	
Gradual Degradation	Light	15 % reduction	Gradual deterioration of structural response
	Moderate	20 % reduction	
	Severe	25 % reduction	
Damping Change	Light	30 % redistribution	Modification of damping around the primary modal frequency
	Moderate	20 % redistribution	
	Severe	10 % redistribution	
Combined Anomalies	Light	+5 % amplitude, 0.14 Hz shift, 10 % damping change	Combined amplitude, frequency, and damping modifications
	Moderate	+7.5 % amplitude, 0.1 Hz shift, 6 % damping change	
	Severe	+12.5 % amplitude, 0.06 Hz shift, 1 % damping change	

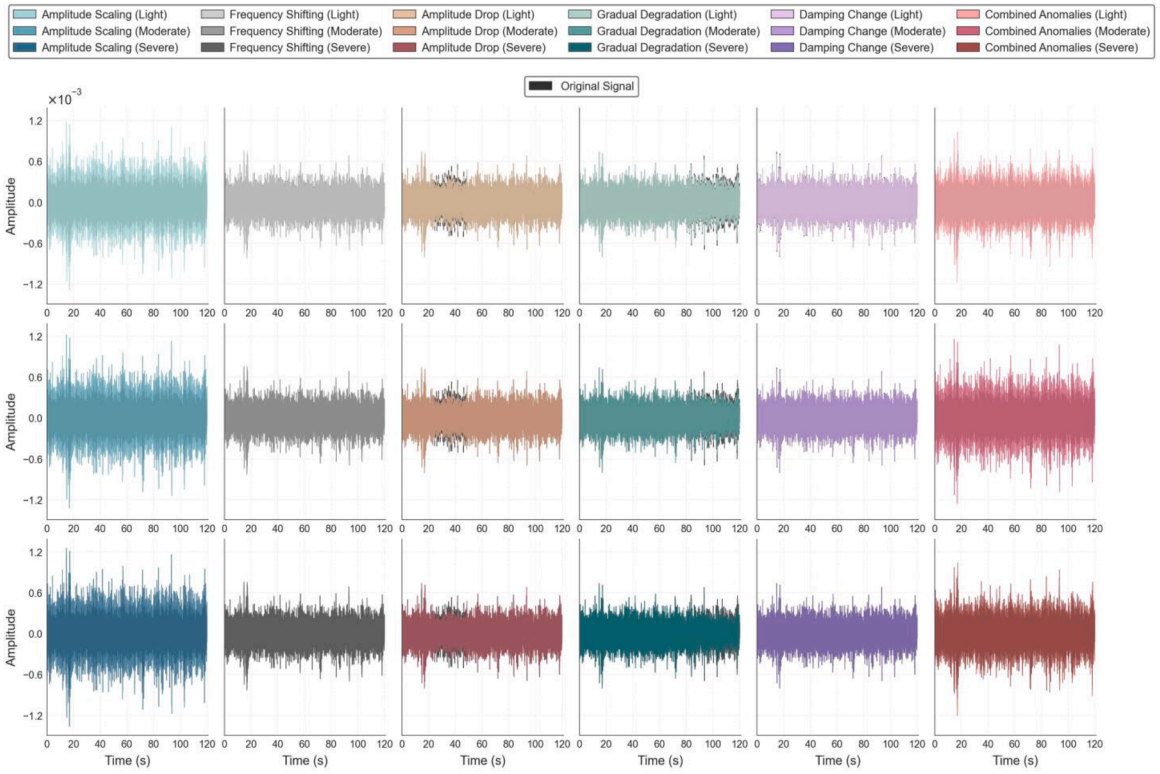


Fig. 11. Simulated anomalies in acceleration signals in three levels for accelerometer A3.

For frequency shifts, a smaller shift (0.1 Hz) represents a severe condition due to its highly localized effect near the primary structural mode (2.34 Hz). A larger shift (0.2 Hz) is classified as light severity due to its more dispersed impact across the frequency spectrum. Similarly, in damping changes, a 10% spectral energy redistribution is considered severe due to its focused effect around the primary mode, while a 30% redistribution represents a light severity due to its broader distribution. This highlights the importance of considering the location and distribution of changes, not just their absolute magnitude, when assessing structural anomalies.

6.6. Evaluation of the Trained MDFA Model

6.6.1. Reconstruction Loss Calculation

The fundamental principle of this unsupervised approach relies on the model's ability to reconstruct healthy signals with high accuracy while exhibiting increased reconstruction errors when presented with anomalous data. Following the training process described in Section 6.3, the MDFA model learned to characterize the bridge's normal operating behaviour using real-world acceleration data. To evaluate its damage detection capabilities, specific anomaly scenarios were defined, as explained in Section 6.5. These scenarios were then introduced to the trained model, and the reconstruction loss was computed. For each accelerometer and anomaly scenario, the model's reconstruction error

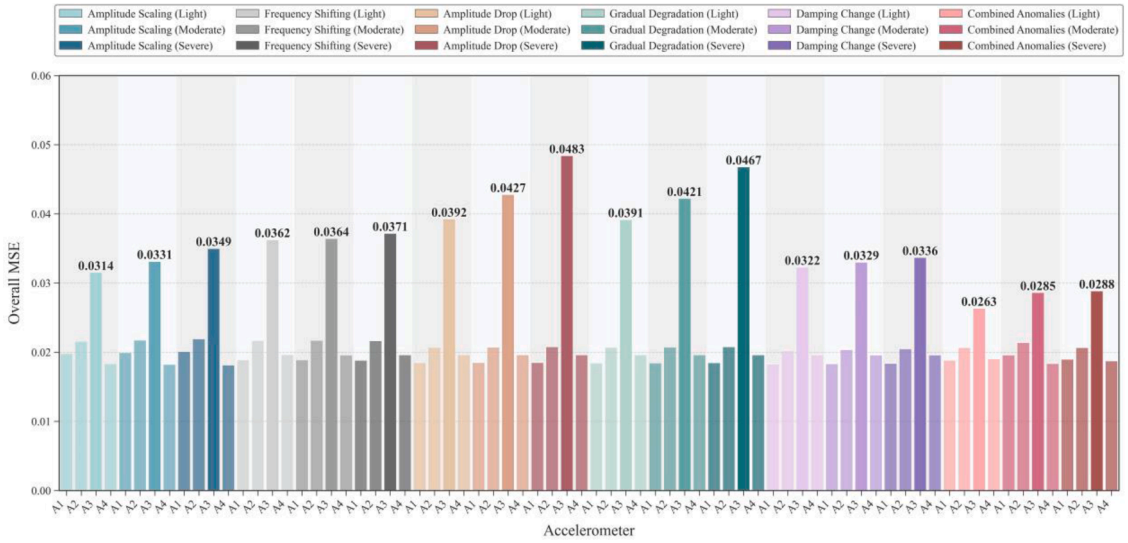


Fig. 12. The MSE values of all accelerometers for each anomaly scenario and severity level.

was quantified using the MSE between the original and reconstructed signals. The overall MSE represents the average of the MSE values calculated across the raw signal reconstruction and the reconstructions of the extracted features in all domains.

Figure 12 presents the MSE values across all accelerometers for each anomaly scenario and severity level. A notable pattern emerges in the reconstruction performance, particularly at accelerometer A3, which was the target location for all simulated anomalies. The MSE values at A3 consistently show elevated levels compared to other accelerometers, demonstrating the model's sensitivity to localized structural changes.

The analysis revealed a strong correlation between anomaly severity and reconstruction error magnitude. As severity levels progressed from light to severe within each anomaly scenario, the corresponding MSE values showed proportional increases. This relationship demonstrates the model's capability to detect not only the presence of anomalies but also their relative intensity.

6.6.2. Feature Importance Analysis

In addition to the MSE analysis, feature importance evaluation was conducted to identify the most significant features for anomaly detection across different damage scenarios. This analysis aimed to optimize the feature extraction process and enhance the sensitivity of the model to subtle changes.

The feature importance was evaluated using a differential MSE approach, which calculates the difference between the reconstruction MSE for damaged states and the baseline MSE from the healthy state. Figure 13 illustrates the feature importance results across different damage scenarios, revealing that Spectral Kurtosis, RMS, and Shannon Entropy are the most important features for detecting anomalies across all anomaly scenarios.

As shown in the figure, these three features demonstrated significantly higher sensitivity to structural changes compared to other extracted features. Their strong performance across various anomaly types confirms their effectiveness in capturing the subtle signal alterations that occur during structural anomalies.

It should be noted that the relative importance of features may vary depending on the type of damage. The current analysis provides insight into feature effectiveness for the specific scenarios considered in this study. These findings enable future optimization of the feature extraction process by prioritizing the most sensitive features, adding new ones, removing less effective features, or replacing them with more sensitive alternatives to enhance the overall performance of the MDFA model.

6.6.3. Results and Discussion

Following the MSE analysis, these values were input into the statistical damage assessment method described in Section 5.3 to compute the damage index. Figure 14 illustrates the damage index values across all accelerometers for different anomaly scenarios and severity levels. The results demonstrate the MDFA model's ability to detect, localize, and quantify the simulated anomaly scenarios. The results show that damage index values at A3, the location where anomalies were simulated, is consistently elevated. This distinct pattern persists across all anomaly types and severity levels, confirming the framework's ability to accurately identify the spatial location of structural changes.

The analysis of damage index values revealed several key findings. Across all anomaly scenarios and severity levels, the damage index for accelerometer A3 significantly exceeded the values observed at other accelerometers, while A1, A2, and A4 maintained consistently low values. This pattern demonstrates the model's ability to accurately detect the presence of anomalies with minimal false positives. Beyond detection, the framework achieved precise damage localization by consistently identifying accelerometer A3 as the anomaly location, enabling targeted inspections and maintenance planning. Additionally, the damage index values at A3 increased proportionally with anomaly severity, from light to severe, demonstrating the model's ability to quantify structural changes effectively. This clear relationship between damage severity and index values provides essential information for maintenance prioritization. Particularly noteworthy was the model's ability to detect subtle structural changes. The framework successfully identified minor variations in frequency shifting and damping characteristics that typically elude visual inspection and conventional signal processing methods. This sensitivity to early-stage anomalies positions the framework as a powerful tool for preventive maintenance.

Consequently, the MDFA-based DT framework addresses a fundamental challenge in SHM by eliminating the need for damage data during training. By maintaining high detection sensitivity and robust performance, the framework proves suitable for practical bridge monitoring applications. The model's comprehensive capabilities in detecting, localizing, and assigning numerical scores that reflect the severity of deviations from normal behaviour support its deployment for real-time monitoring of the Werrington bridge. This evaluation framework represents a significant advancement in SHM methodology. Traditional approaches often require either destructive testing of actual structures or reliance on simplified laboratory models. Our framework circumvents these limitations through a two-stage process. First, the TTS-WGAN-GP model was employed to generate synthetic acceleration data that closely mirrors the bridge's normal operating conditions. Subsequently, various anomaly scenarios were applied to these synthetic signals to simulate structural damage. These modifications were not intended to imply localized physical damage, but rather to emulate abnormal dynamic responses characteristic of structural anomalies. These controlled perturbations reflect anomaly patterns that may arise from various types of

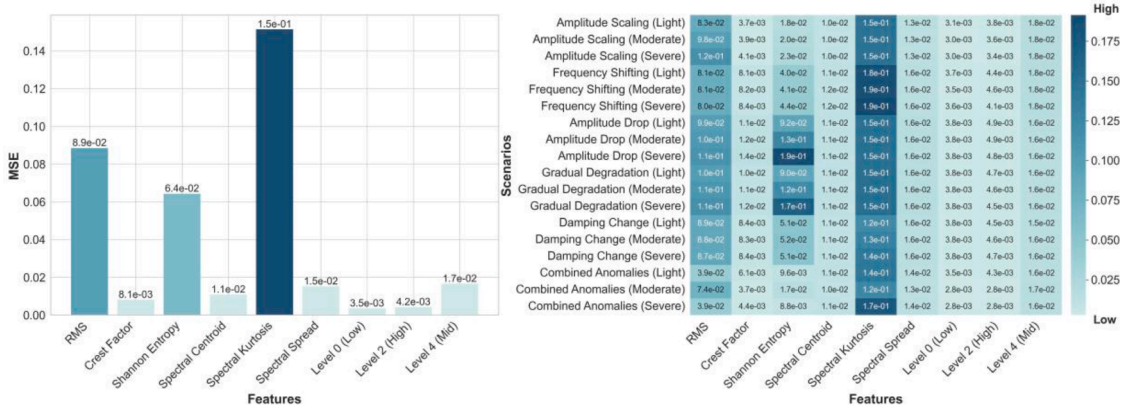


Fig. 13. The feature importance results (a): Relative importance of features based on differential MSE (b): Heat map of feature importance across damage scenarios.

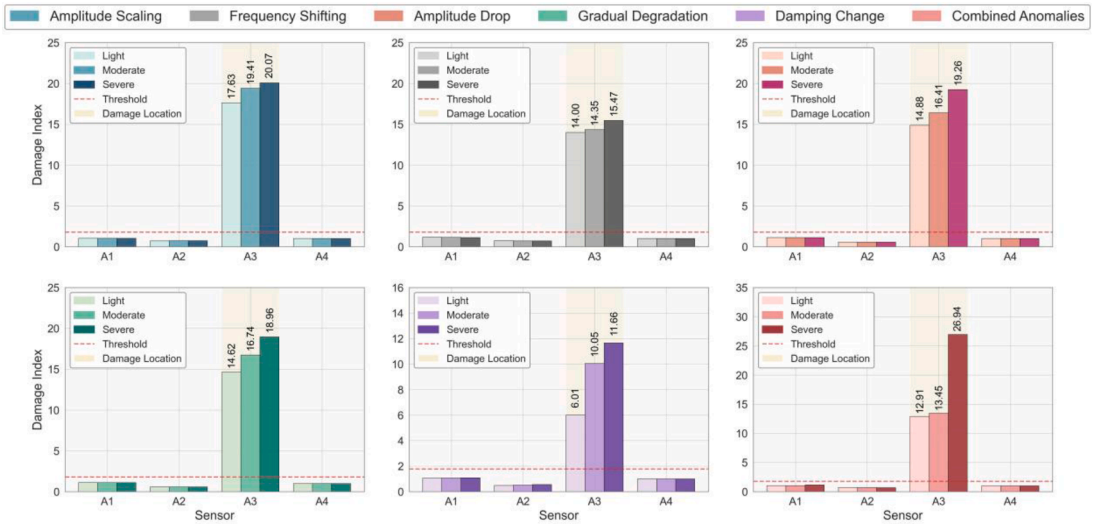


Fig. 14. The damage index values for all accelerometers for different anomaly scenarios and severity levels.

structural degradation and facilitate the evaluation of the model's sensitivity and performance, thereby providing an effective proof of concept for anomaly detection. This approach paves the way for more efficient and practical development and deployment of advanced SHM systems for civil infrastructure.

7. Comparison with Existing Damage Detection Method

In this section, the proposed MDFA was compared with a multi-head self-attention long short-term memory auto encoder (MA-LSTM-AE) to evaluate their respective performances in detecting simulated anomaly scenarios on the Werrington bridge data. The MA-LSTM-AE model, introduced in [25], successfully detected all damage scenarios in three separate case studies, including an eight-degrees-of-freedom mass--spring system, a three-story experimental frame structure, and the full-scale Z24 bridge. However, the following comparison shows that, under the subtle anomalies simulated in this study, the MDFA demonstrated superior sensitivity and detection capability.

Table 6 presents the damage index values at Sensor A3 for both models across six different anomaly types and three severity levels. The same simulated abnormal scenarios, as described at Section 6.4, used to test the ability of each model to detect and quantify subtle changes in the acceleration signals.

Table 6
Comparison of damage index values at Sensor A3 under various anomaly scenarios between the MDFA and MA-LSTM-AE models.

Scenario	Severity	MDFA	MA-LSTM-AE
Amplitude Scaling	Light	17.63	1.081
	Moderate	19.41	1.216
	Severe	20.07	1.361
Frequency Shifting	Light	14.00	0.002
	Moderate	14.35	0.001
	Severe	15.47	0.003
Amplitude Drop	Light	14.88	0.040
	Moderate	16.41	0.055
	Severe	19.26	0.071
Gradual Degradation	Light	14.62	0.050
	Moderate	16.74	0.067
	Severe	18.96	0.085
Damping Change	Light	06.01	0.001
	Moderate	10.05	0.000
	Severe	11.66	0.000
Combined Anomalies	Light	12.91	0.457
	Moderate	13.45	1.032
	Severe	26.94	0.671

The results indicate that the MDFA consistently produced higher damage index values, even when the anomalies were minor. By contrast, the MA-LSTM-AE failed to capture these subtle changes, as reflected by the relatively low damage indices across most scenarios, especially in frequency shifting and damping change scenarios. This outcome highlights the differences between the two approaches, particularly regarding how additional feature representations enhance the detection process.

The MDFA's improved performance is largely attributed to its feature fusion mechanism, which integrates information from the raw acceleration signals and domain-specific features in the time, frequency, and wavelet domains. This approach allows the model to exploit complementary information that might be overlooked when relying solely on raw signals. The cross-modal attention blocks, designed to learn correlations among different domain representations, enable the MDFA to capture complex relationships in the structural response. By weighting the importance of each domain representation, the model becomes more sensitive to minor anomalies that manifest across multiple domains, thereby increasing its detection accuracy.

While the MA-LSTM-AE has proven effective in detecting damage in various laboratory experiments and a full-scale bridge, the subtle nature of the anomalies in this study presented a greater challenge. The results suggest that including domain-specific features and implementing a cross-modal attention mechanism can significantly improve damage detection performance in scenarios where structural changes are difficult to discern. Consequently, the MDFA model's ability to integrate raw and domain-transformed features positions it as a powerful and flexible approach for real-time, unsupervised SHM applications.

8. Case Study II: The S101 Bridge

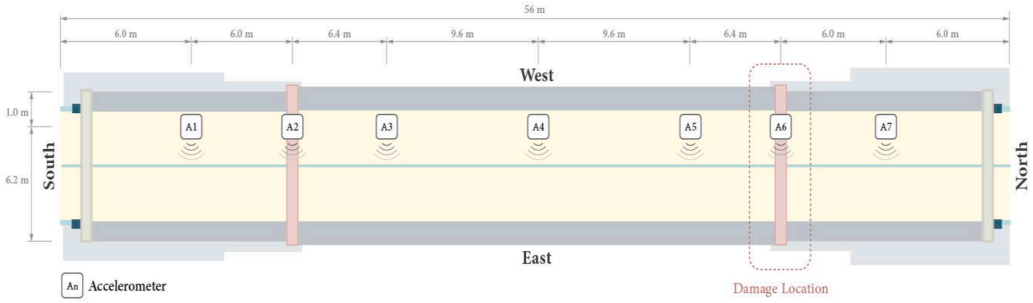
8.1. Dataset

For the second case study, the S101 Bridge, as shown in Figure 15(a), was selected which is a prestressed concrete highway bridge located near Vienna in Reibersdorf, Austria. Constructed in the early 1960s, the bridge featured a three-span configuration with a 32-meter main span and two 12-meter side spans. By 2008, the bridge had experienced significant structural deterioration and insufficient load-bearing capacity, leading to its scheduled demolition [59].

Prior to demolition, the bridge became the site of a comprehensive progressive damage test conducted from December 10 to 13, 2008. The primary objective of this test was to gather high-quality data for advancing SHM methodologies through carefully controlled structural



(a)



(b)

Fig. 15. (a): The S101 Bridge; (b): Accelerometers placement and damage locations.

damage scenarios. During the testing period, the bridge was closed to vehicular traffic, with excitations primarily arising from ambient vibrations, while one lane of the highway below remained operational [60].

The bridge was subjected to two distinct forms of controlled damage. The first involved deliberately cutting and gradually lowering the north-western pier to simulate settlement effects. The second phase consisted of systematically severing tendons to replicate the loss of prestress in the structure. In this paper, only results obtained using data relevant to the settlement of the pier are reported. A more detailed description of these progressive damage scenarios is reported in Table 7.

To monitor the bridge's dynamic behaviour, 15 triaxle acceleration sensors were strategically deployed across the structure, 14 along the east side of the deck and one on the west side to capture torsional modes. These sensors recorded data continuously at a sampling frequency of 500 Hz in 5.5-minute sets, generating 714 data files, each containing 165,000 data points. The accelerometers remained active throughout the testing period, recording the bridge's response before, during, and after each damage step.

For computational efficiency in this study, data from only 7 of the 15 accelerometers was utilized, the locations of which are shown in Figure 15(b). Although accelerations in three orthogonal directions were recorded by the sensors, only the vertical acceleration component was used in this study. Furthermore, the analysis was limited to 11 minutes of data for each scenario (both training data and damage scenarios), which corresponds to 330,000 samples per scenario.

Table 7
Description of progressive damage scenarios for S101 bridge.

Scenario	Damage Location	Description
1	-	Undamaged structure
2	A6	Cut of the north-western column
3	A6	Lowering the column (1 cm)
4	A6	Lowering the column (2 cm)
5	A6	Lowering the column (3 cm)
6	A6	Insertion of compensating plates

8.2. Evaluation Result and Discussion

Following the evaluation of the proposed model on simulated anomaly scenarios in the first case study, the same model configuration was applied to real-world damage scenarios from the S101 bridge without any modifications to the hyper-parameters. This approach was intended to assess whether a model that successfully detected subtle simulated anomalies could also detect actual damage under more complex and less controlled conditions.

As shown in Figure 16, the MDFA model successfully identified and localized all damage scenarios. In Scenario 1, which represents the healthy state of the bridge, the damage indices for all sensors remained below the threshold. By contrast, in the subsequent scenarios involving the cutting and lowering of the north-western pier, Sensor A6 exhibited significantly higher damage indices than the other sensors, reflecting its proximity to the damaged area. Minor increases in the damage indices of the remaining sensors were also observed, likely due to changes in the overall dynamic behaviour of the bridge as a result of the localized damage.

The model's estimation of relative damage severity was generally consistent with the nature of the damage. Specifically, lowering the pier produced a higher damage index than cutting the pier, suggesting that the model captured the more pronounced impact of pier settlement on the bridge's dynamic characteristics. However, the model did not show a clear incremental increase in the damage index when the pier was lowered progressively from 1 cm to 3 cm. This outcome may be attributed to the limited training data and the presence of environmental and operational variations during the damage application, both of which were not fully represented in the 11 minutes of healthy-state data used to train the model.

Despite these constraints, the model demonstrated considerable sensitivity to structural changes. The damage indices showed a clear distinction between the undamaged and damaged states, even though only a small amount of training data was available and the model's hyper-parameters were not optimized for this specific bridge. This sensitivity, while beneficial for detecting damage, can also lead to higher damage indices under subtle shifts in environmental or operational

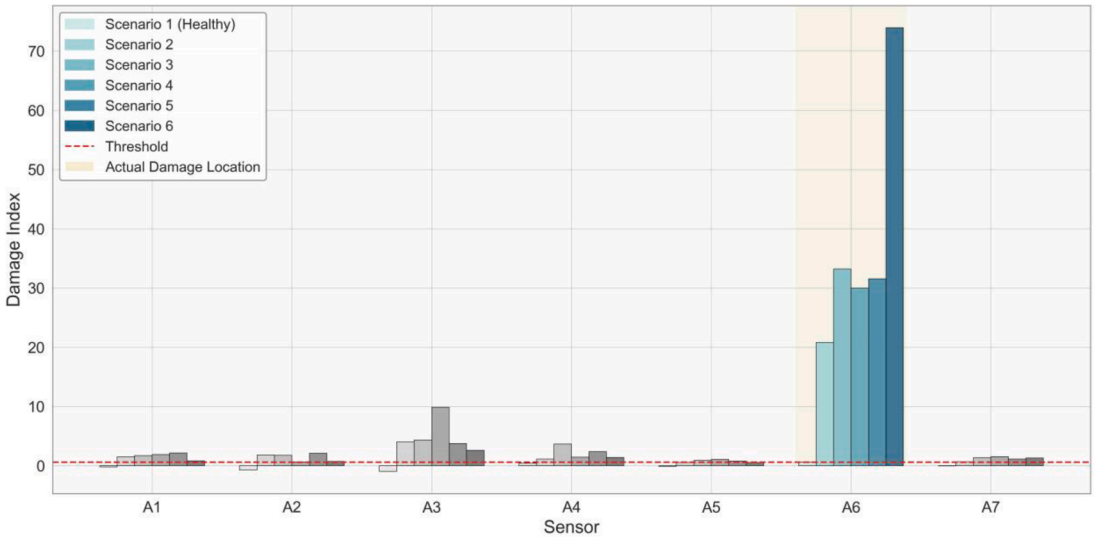


Fig. 16. The results of MDFA model on damage scenarios of S101 bridge.

conditions. Nevertheless, the results suggest that simulated anomaly scenarios can serve as an effective means of evaluating unsupervised damage detection methods when real-world damage data are scarce or difficult to obtain

It is anticipated that the model's robustness could be further improved by incorporating additional sources of data, such as temperature, wind, and traffic loads, to better capture the range of variations encountered by the structure. Optimizing the hyper-parameters and refining the feature extraction process could also enhance the accuracy and stability of the damage detection results. Overall, the model's performance in this case study indicates that it is capable of detecting real-world damage with minimal training data and no specialized parameter tuning, offering a promising avenue for future research in SHM.

9. Conclusions and Future Works

This research introduces an advanced data-driven DT framework for real-time bridge damage detection, integrating feature fusion with ML methodologies. The proposed DT employs a Multi-Domain Fusion Auto encoder (MDFA), facilitating a comprehensive analysis of structural responses. This innovative methodology enhances bridge management by enabling the development of early warning systems for predicting safety risks. By monitoring structural responses in real time, the proposed approach supports proactive risk mitigation strategies, strengthening the safety and resilience of bridge infrastructure.

The effectiveness of the proposed method was assessed through two distinct evaluation approaches. Initially, a novel evaluation strategy was proposed, where abnormal signals were simulated on Werrington bridge data, and the MDFA model was assessed using this approach. The second evaluation was conducted on the S101 Bridge in Austria, using real-world damage scenarios to validate the method's performance in practical situation. A comparative analysis was also performed against a well-performing unsupervised MA-LSTM-AE method to evaluate the effectiveness and advantages of the proposed approach.

The key findings of this study can be summarized as follows:

1. The evaluation results of the proposed data-driven DT framework in simulated scenarios in Werrington bridge data demonstrated successful detection of subtle variations in frequency shifting and damping characteristics, which are often undetected in traditional signal processing methods.
2. The comparison results of the proposed framework against MA-LSTM-AE method showed significantly superior results, with higher accuracy in detecting structural changes. The MA-LSTM-AE model struggled to identify changes, particularly in scenarios involving frequency shifts and damping variations. These findings highlight the superior performance of the proposed MDFA model in detecting early-stage and low-severity damage.
3. The superior results of the MDFA model underscore the importance of feature fusion techniques in improving the accuracy of real-time damage detection within the data-driven DT framework. Furthermore, the modularity of the proposed DT framework allows for the integration of environmental factors such as temperature, wind, and traffic, improving the reliability of results and increasing the model's robustness against external factors.
4. The novel evaluation strategy proposed in this study addresses a critical gap in assessing unsupervised learning models for SHM. By using transformer-based GANs to generate realistic synthetic anomalies on real bridge data, this approach overcomes the limitations of traditional evaluation methods that rely on either scarce historical damage data or simplified laboratory conditions that fail to capture real-world complexity.
5. The results of the MDFA model on real damage data from the S101 bridge shows successfully identified all damage scenarios within 11 minutes of data collection, despite varying environmental conditions, without requiring any additional optimization step. This demonstrates the effectiveness of the proposed evaluation strategy based on synthetic anomaly simulations, which can serve as a benchmark for laboratory and scaled structures, providing a cost-effective and time-saving approach to assessing unsupervised damage detection methods.

The key findings of this study highlight the effectiveness of the proposed approach in advancing bridge monitoring and management, leading to enhanced safety and operational efficiency in transportation network.

However, this study did not incorporate an optimization process for the proposed DT framework and MDFA model. Future work could focus on enhancing performance by refining feature extraction, tuning hyper-parameters, and adjusting the model architecture to improve both training efficiency and damage detection accuracy. Additionally, the training time and resource usage for the current model can be improved by development of lighter model variants to further enhance its scalability. Furthermore, testing the performance of the model in real time scenarios on edge or embedded devices and optimizing the model for real-time responsiveness will be another key focus in future research.

In addition, the high sensitivity of the MDFA model to signal variations may lead to errors caused by environmental influences. This study did not include environmental factors such as temperature, traffic, and wind into the proposed model. Future research may explore the integration of these factors to enhance the model's robustness. Additionally, extending the framework to support multi-sensor fusion and more validation results across diverse bridge types with more complex and physically damage scenarios could be further studied in future works. Another area for further research could be the extension of the proposed framework to incorporate real time Decision Support System (DSS) for bridge management, with a particular focus on optimal bridge maintenance planning. Building upon the methodologies introduced in this study, researchers can develop real time DSS techniques to assess structural deterioration and optimise the maintenance plans for long-term infrastructure management.

List of abbreviations

AI	Artificial Intelligence
1D CNN	1-dimensional Conventional Neural Network
BMS	Bridge Management Systems
CNN-BiLSTM-MHA	Convolutional Neural Network–Bidirectional Long Short-Term Memory–Multi-Head Attention
DL	Deep Learning
DTs	Digital Twins
DSF	Damage-Sensitive Features
DSS	Decision Support System
FEM	Finite Element Method
FID	Fréchet Inception Distance
GAN	Generative Adversarial Networks
IoT	Internet of Things
MDFA	Multi-Domain Fusion Autoencoder
ML	Machine Learning
MSE	Mean Squared Error
PDF	Probability Density Function
PSD	Power Spectral Density
RMS	Root Mean Square
SHM	Structural Health Monitoring
SVM	Support Vector Machine
TTS-WGAN-GP	Transformer-based Time-Series Wasserstein Generative Adversarial Network with Gradient Penalty

CRediT authorship contribution statement

Vahid Mousavi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria Rashidi:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Shayan Ghazimoghadam:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bijan Samali:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The code and data utilized in this study are available upon request from the Corresponding Author.

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